ACTIVITY 1: GENE PROFILE INVESTIGATION

Cell and Molecular Biology Short-Term Assignment

Name:	Emperado, Faith Denielle	**Course/Section:**	BIO - 300 - A
Assigned gene:	CD19	**Due date:**	07/21/2026
Partner (if any):		**Instructor:**	Ma’am, Lilibeth Bucol

Purpose

In this activity, you will investigate one gene using the National Center for Biotechnology Information (NCBI) databases. You will learn how to locate and interpret basic information about a gene, including its official name, chromosomal location, gene structure, protein product, biological function, and association with disease or traits.

Part A. Find the Correct Gene Record

1. Open the NCBI Gene website using the link above.
2. In the search box, type the gene symbol or gene name followed by the organism. Example: TP53 Homo sapiens.
3. Select the record that clearly states “Homo sapiens (human).” Do not select a record from another species.
4. At the top of the record, confirm the official gene symbol, official full name, organism, and Gene ID.
5. Copy the webpage address of the gene record for your references.

Record the Following Information

Item	Your Answer
Official gene symbol	CD19
Official full gene name	CD10 molecule
Organism	*Homo sapiens*
NCBI Gene ID	930
Gene type (for example, protein-coding)	Protein-coding
NCBI Gene record URL	https://www.ncbi.nlm.nih.gov/datasets/gene/930/

Checkpoint: A gene symbol is usually short and written in capital letters for human genes, such as TP53. The Gene ID is a unique number assigned by NCBI. Do not confuse the Gene ID with a transcript or protein accession number.

Part B. Determine the Chromosomal and Genomic Location

1. In the NCBI Gene record, locate the "Genomic context" section.

2. Record the chromosome number and the cytogenetic location, when available. A cytogenetic location may appear in a form such as 17p13.1.

3. Record the genomic coordinates shown for the current reference genome assembly. Include the assembly name if it is displayed.

4. Note the strand orientation. The gene may be located on the plus strand or minus strand.

5. Write one sentence explaining what the chromosome location means.

Item	Your Answer
Chromosome	16
Cytogenetic location	16p11.2
Genome assembly	GRCh38.p14
Genomic coordinates	NC_000016.10: 28931971-28939342 (+)
Strand orientation	Positive strand (+)
One-sentence interpretation	This means that the CD19 gene is found on chromosome 16 at cytogenetic band 16p11.2, where it occupies a specific region of DNA on the positive strand of the human genome (GRCh38.p14).

Part C. Examine Gene Structure

1. From the NCBI Gene record, open the Genome Data Viewer link, or search the assigned gene directly in Genome Data Viewer.

2. Make sure the organism is Homo sapiens and that the correct gene is selected.

3. Zoom in until you can see the gene model and the exon-intron structure.

4. Identify the direction of transcription from the arrows or strand information.

5. Count the exons in the selected reference transcript. Use one clearly labeled RefSeq transcript, preferably an NM_ accession for a protein-coding human gene.

6. Record the transcript accession. Then identify the corresponding protein accession, preferably an NP_ accession, from the NCBI Gene record or RefSeq section.

7. Take one screenshot showing the gene model. Your screenshot must include the gene name and visible exon structure.

Item	**Your Answer**
Reference transcript accession	NM_001770.6
Number of exons in that transcript	15
Protein accession	NP_001761.3
Direction of transcription	Towards the right
Screenshot file name	CD19_genomedataviewer.screenshot.png

Important: Different transcript variants may have different numbers of exons. Therefore, always report the accession number of the transcript you examined. Do not report an exon count without naming the transcript.

Part D. Investigate the Protein Product

1. Locate the “RefSeq” or “NCBI Reference Sequences” section of the Gene record.

2. Open the selected protein record linked to the reference transcript.

3. Record the protein name and amino-acid length.

4. Identify the main cellular location of the protein, when stated. Examples include nucleus, cytoplasm, plasma membrane, mitochondrion, or extracellular space.

5. Identify one important domain, motif, or functional region, if available in the protein record or conserved-domain section.

Item	**Your Answer**
Protein name	B-lymphocyte antigen CD19 isoform 2 precursor
Protein accession	NP_001761.3
Protein length (amino acids)	556 amino acids
Main cellular location	Plasma membrane
One domain, motif, or functional region	Two extracellular immunoglobin (ig)

Part E. Explain the Biological Function

- The CD19 molecule, a transmembrane glycoprotein that is mainly expressed on the surface of B cells, is encoded by the CD19 gene. As a crucial co-receptor, CD19 improves signaling via the B-cell receptor (BCR), enabling B cells to identify antigens and react appropriately. By attracting intracellular signaling molecules following antigen recognition, it takes part in B-cell activation, proliferation, differentiation, and antibody production. A crucial part of the BCR signaling pathway, CD19 lowers the threshold needed for B-cell activation, which aids in controlling immune responses. In order for the adaptive immune system to produce effective antibody-mediated immunity while maintaining appropriate regulation of immune responses, normal CD19 function is required for the development and maintenance of healthy B cells.

Part F. Gene and Disease or Phenotype

- One uncommon primary immunodeficiency disorder linked to the CD19 gene is CD19 deficiency. Pathogenic variations in the CD19 gene cause B cells to express less or no CD19 protein, which disrupts B-cell receptor signaling and interferes with the normal production of antibodies. Because their immune systems are unable to generate sufficient protective antibodies, afflicted people frequently suffer from recurrent bacterial infections, especially of the respiratory tract. Although CD19 deficiency is directly caused by pathogenic CD19 mutations, altered CD19 expression has also been linked to some B-cell leukemias and lymphomas, where changes in expression may impact the course of the disease or the response to immunotherapy. Meaning, some CD19 abnormalities are causative, while others are linked to clinical outcomes or disease risk rather than being the direct cause.

Reference

National Center for Biotechnology Information. NCBI Gene: CD19 molecule (*Homo sapiens*), Accessed July 22, 2026. https://www.ncbi.nlm.nih.gov/gene/930

National Center for Biotechnology Information. RefSeq Protein NP_001761.3 (CD19 molecule). Accessed July 22, 2026. https://www.ncbi.nlm.nih.gov/protein/NP_001761.3

https://www.ncbi.nlm.nih.gov/genome/gdv/

Appendix

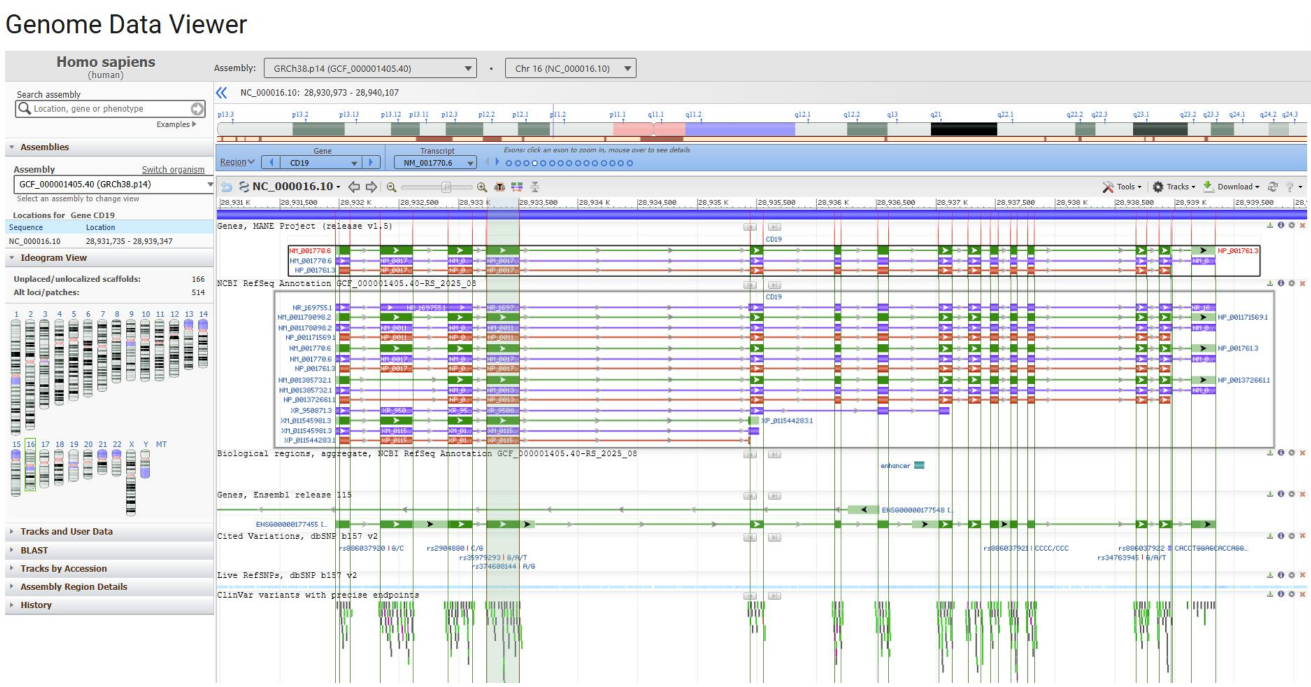

FIGURE 1. **Genome Data Viewer showing the genomic structure of the human CD19 gene on chromosome 16p11.2.** The figure displays the RefSeq transcript NM_001770.6, exon–intron organization, direction of transcription on the positive (+) strand, and genomic coordinates in the GRCh38.p14 human reference genome.